

Pattern Recognition

We live in an age of data abundance. Data of various types such as imagery, text, graphs, tabular, time series, and more are being collected on a daily basis across a wide range of societal settings. This data has hidden information that can be leveraged to improve problem solving, inform decision making, make scientific discoveries, and many other possibilities. But how do we find and extract information from data? Pattern recognition is the scientific discipline whose goal is the classification of objects into a number of categories and classes. Pattern recognition is an integral part of most machine intelligence systems built for decision making, and provides essential tools for analyzing and understanding data.

This course will give an in-depth introduction to the discipline of pattern recognition, and provide students with a wide range of essential tools that any machine learning practitioner should have in their tool belt.

Overview of course

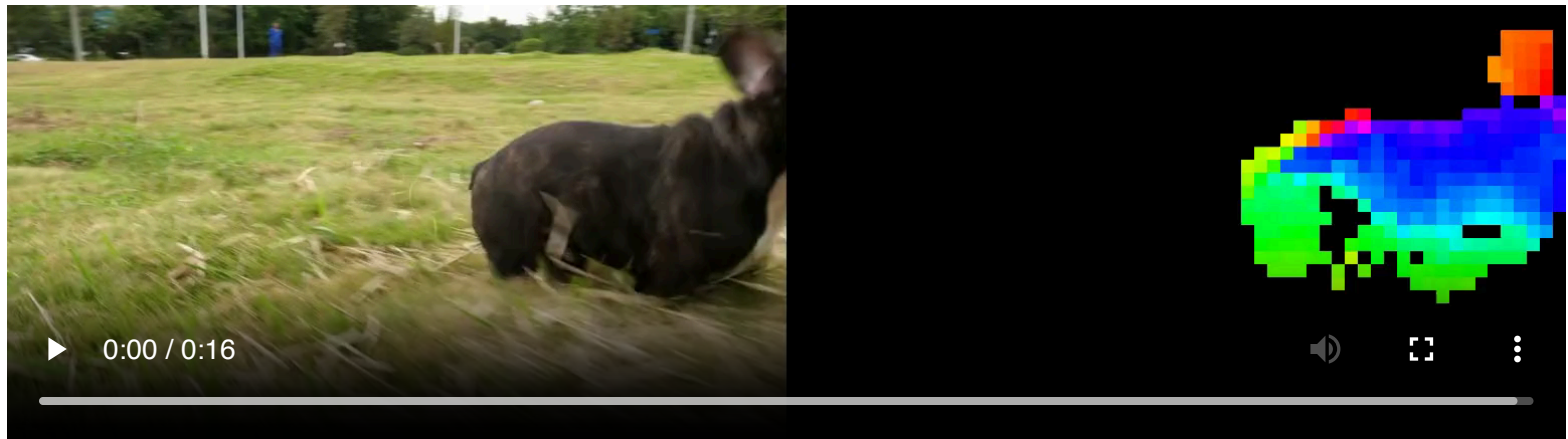
- The course into 5 main components:
 - Bayes decision theory.
 - Linear classifiers.
 - Non-linear classifiers.
 - Data transformations.
 - Clustering.
- Content for each component is contained in notebooks. The notebooks are presented in class as slides made with [Marimo](#) and hosted with [GitHub Pages](#). For this reason, the text in the notebooks is organized with bullet points.

What will you have at the end of this course?

1. You will have a deep understanding of the fundamental techniques and algorithms that are frequently used in practice:
 - "The first thing you try"-techniques like linear classifiers, k-means clustering, and principal component analysis will be covered in-depth and implemented from scratch.
2. You will learn how to think more theoretically about how to approach a data analysis task.
 - Detailed derivations of complex topics like kernel methods, neural networks, and Laplacian Eigenmaps.
3. You will derive state-of-the-art algorithms from first principles.
 - Derivation of widely used algorithms like Support Vector Machines.
4. You will have hands-on experience from working with diverse data and tasks.

Applications of pattern recognition - PCA in computer vision

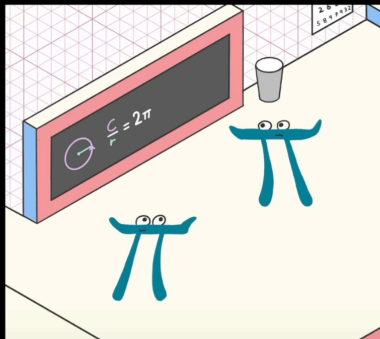
Video from DINOv2 repository



Applications of pattern recognition - Temperature scaled softmax in GPT

Image from 3Blue1Brown.

Behold, a wild pi creature,
foraging in its native _____



land	22%
forest	9%
country	5%
habitat	4%
forests	4%
soil	4%
territory	2%
woods	2%
lands	1%
waters	1%
woodland	1%
grass	1%
⋮	

Applications of pattern recognition - Bayes decision theory in image generation models

Image from GroundingDino and from Lil'log

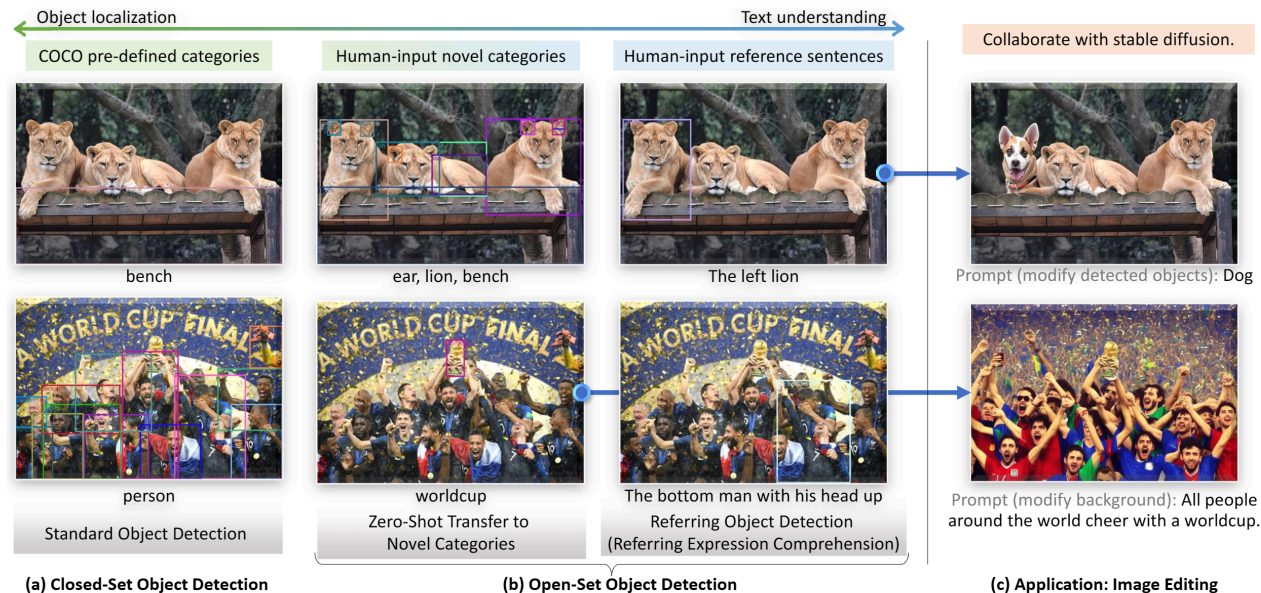


Figure 1. (a) Closed-set object detection requires models to detect objects of pre-defined categories. (b) Previous work zero-shot transfer models to novel categories for model generalization. We propose to add Referring expression comprehension (REC) as another evaluation for model generalizations on novel objects with attributes. (c) We present an image editing application by combining Grounding DINO and Stable Diffusion [42]. Best view in colors.

Using Bayes' rule, we have:

$$\begin{aligned}
 q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \\
 &\propto \exp\left(-\frac{1}{2}\left(\frac{(\mathbf{x}_t - \sqrt{\alpha_t}\mathbf{x}_{t-1})^2}{\beta_t} + \frac{(\mathbf{x}_{t-1} - \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0)^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{1 - \bar{\alpha}_t}\right)\right) \\
 &= \exp\left(-\frac{1}{2}\left(\frac{\mathbf{x}_t^2 - 2\sqrt{\alpha_t}\mathbf{x}_t\mathbf{x}_{t-1} + \alpha_t\mathbf{x}_{t-1}^2}{\beta_t} + \frac{\mathbf{x}_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0\mathbf{x}_{t-1} + \bar{\alpha}_{t-1}\mathbf{x}_0^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{1 - \bar{\alpha}_t}\right)\right) \\
 &= \exp\left(-\frac{1}{2}\left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}}\right)\mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t}\mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}}\mathbf{x}_0\right)\mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0)\right)\right)
 \end{aligned}$$

$$-\log p_\theta(\mathbf{x}_0) \leq -\log p_\theta(\mathbf{x}_0) + D_{\text{KL}}(q(\mathbf{x}_{1:T}|\mathbf{x}_0) \| p_\theta(\mathbf{x}_{1:T}|\mathbf{x}_0))$$

University courses

These notebooks are currently adopted in [FYS-3012 Pattern Recognition](#) at UiT the Arctic University of Tromsø.

Roadmap

This is an early version of the course. There might be imprecisions and errors. Also, some chapters might undergo significant revisions and changes.

I am planning to add more content over time to cover additional topics in new chapters and to extend the existing ones with new examples and explanations. If there is a specific topic you feel is missing or passages that you feel are not clear enough, open an [Issue](#) on the repository on Github.

A note on deep learning

This course intentionally avoids advanced deep learning topics. This is mainly to not overlap with the dedicated deep learning course at UiT [FYS-3033 Deep Learning](#). However, the foundation of deep learning (neural networks) is introduced from first principles. We go all the way from the Perceptron to multilayer perceptrons, coding everything from scratch. That means, those who follow this course will have a fundamental understanding of how neural networks work. Then, the dedicated deep learning course expands on this to cover the fundamental and most recent topics within the field of deep learning.

Practical information for 2026 version of UiT course

Teaching staff

1. Associate professor Kristoffer Wickstrøm - course responsible
2. Associate professor Veronica Lachi - co-leader of the course
3. PhD candidate Riccardo Gelato - teaching assistant

Lecture and exercises

1. Lectures:

- Fridays from 08:15 - 10:00 in FPARK Klasserom 2 (B284)
- Mondays from 10:15 - 12:00 in FPARK Klasserom 2 (B284)

2. Exercises:

- Thursdays from 10:15 - 12:00 in TEKNOBYGGET Rom 1.017, Undervisningsrom

Exam

1. Home exam counting 50% of grade:
 - Starts in week 40, handed in in week 42.
2. School exam in week 49 counting 50% of grade.

On the use of LLM and AI agents

LLMs and AI agents are fantastic tools. They can make projects more fun and engaging, and can allow you to take your projects much further than you could otherwise. But remember, your solution to exercises and exams is not the products, you are! If you want to understand pattern recognition, you need to be the one putting in the work, implementing algorithms and solving exercises.

Interesting resources on the topic of AI and its potential impact:

- [What will be left for us to work](#)
- [AI as Normal Technology](#)